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Department of Management Studies
Navi Mumbai

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TITLE PAGE

- **Theme:** Assistive Technology and Inclusive Education
- **Problem Statement Title:** Empowering Differently Abled Students Through Smart Assistive Technologies
- **Idea Title: SignLens EDU**
- **Team ID: Received on email**
- **Team Name: Silent Coders**

SignLens EDU

List of Feature Offered by the Solution

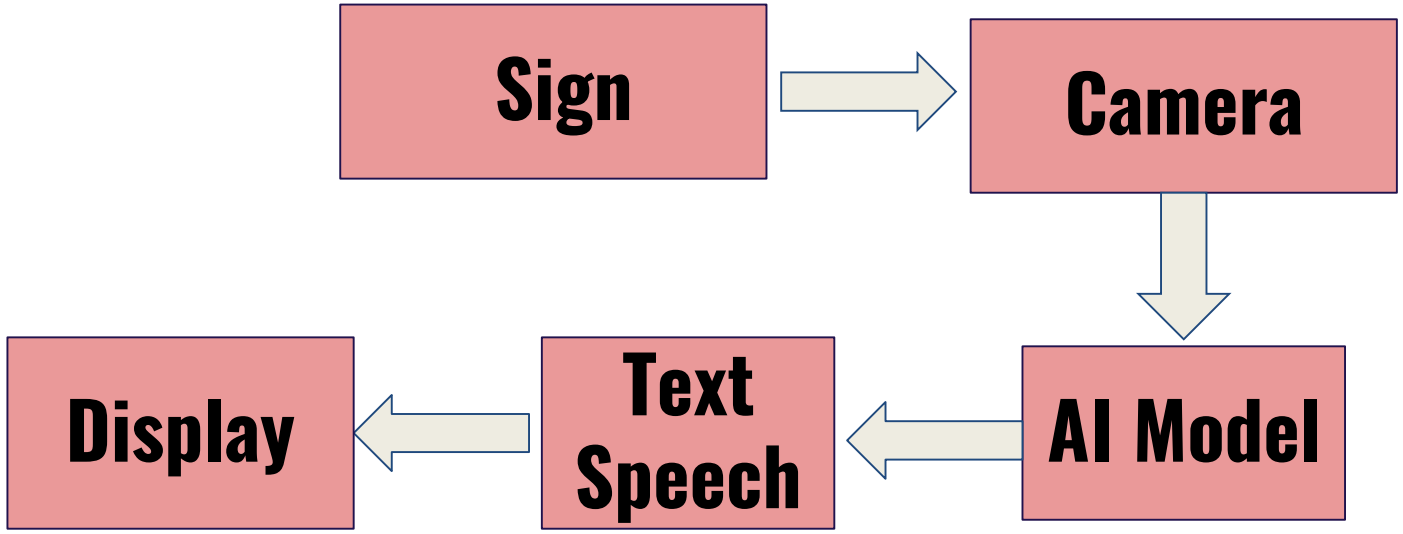
- **AI-powered Interpreter:** Harnesses advanced AI for accurate,real-time sign language translation.
- **Real-time Translation:** Converts sign Language to text and speech instantly,and vice versa.
- **Browser-based :** Accessible on most devices,requiring minimal setup.
- **Personalized Learning:** Supports individual student needs,enhancing engagement.
- **Seamless Teacher Integration:** Intuitive design that allows educators to use the tool effortlessly during classroom teaching.

TECHNICAL APPROACH

◆ Technologies Used

- MediaPipe (Hand Tracking)
- TensorFlow (Gesture Classification)
- HTML, CSS, JavaScript (Frontend)
- Web Speech API (TTS + STT)
- Python Flask (Backend)
- Firebase / MySQL (Data Storage)

◆Methodology



FEASIBILITY AND VIABILITY

❑ FEASIBILITY AND VIABILITY

1. Scalable Design: Can easily expand to more signs, mobile app, offline mode & school integration.
2. Low Resource Requirements: Works with a normal webcam and browser — no special hardware.
3. Uses Proven Technologies: Built using reliable tools like MediaPipe, TensorFlow, Flask, and Web Speech API.

❑ CHALLENGES & RISKS

1. Variations in Signing Styles: Different speeds, hand shapes, and individual signing differences.
2. Lighting/Camera Issues: Poor lighting or incorrect angles may reduce accuracy.
3. Limited ISL Data

◆ Strategies for overcoming these challenges

We will overcome these challenges by training on diverse signer data with augmentation, adding on-screen guidance for lighting and camera setup, and expanding ISL datasets through custom sample collection and crowd-sourced data.

IMPACT AND BENEFITS

◆ Impact on Target Audience

- ➔ Deaf students actively **participate in classrooms**
- ➔ Improved teacher–student **communication**
- ➔ Builds **confidence** & **reduces** social isolation

◆ Social Benefits

- ➔ Promotes inclusion and **equal communication**
- ➔ Reduces **dependency on interpreters**
- ➔ Helps students **express themselves freely**

◆ Technological Benefits

- ➔ Uses **open-source AI tools**
- ➔ **Browser-based**, easy to scale and upgrade
- ➔ Supports future expansion **(mobile app, offline mode)**

RESEARCH AND REFERENCES

◆References

1. WHO, "Deafness and Hearing Loss", (https://www.who.int/news-room/fact-sheets/detail/deafness-and-hearing-loss)
2. 2. Indian Sign Language Research and Training Centre (https://islrtc.nic.in/)
3. 3. World Federation of the Deaf (https://wfdeaf.org/unesco-1880/)

◆Research Tools

1. 1. Technical Foundation for Gesture Tracking Google AI
2. (https://developers.google.com/mediapipe/solutions/vision/hand_landmarker)
3. 2. Web Speech API (MDN), (https://developer.mozilla.org/en-US/docs/Web/API/Web_Speech_API)
3. Google Teachable Machine (https://teachablemachine.withgoogle.com/)